

Recap lecture 21

Example of Moore machine, Mealy machine, Examples, complementing machine, Incrementing machine.

Applications of Incrementing and Complementing machines

1's complementing and incrementing machines which are basically Mealy machines are very much helpful in computing.

The incrementing machine helps in building a machine that can perform the addition of binary numbers.

Using the complementing machine along with incrementing machine, one can build a machine that can perform the subtraction of binary numbers, as shown in the following method

Subtracting a binary number from another

Method: To subtract a binary b from a binary number a

1. Add 1's complement of b to a (ignoring the overflow, if any)
2. Increase the result, in magnitude, by 1 (use the incrementing machine). Ignoring the overflow if any.

Note: If there is no overflow in (1). Take 1's complement once again in (2), instead. This situation occurs when b is greater than a , in magnitude. Following is an example of subtraction of binary numbers

Example

To subtract the binary number 101 from the binary number 1110, let

$a = 1110$ and $b = 101 = 0101$.

(Here the number of digits of b are equated with that of a)

i) Adding 1's complement (1010) of b to a.

1110

+1010

11000 which gives 1000 (ignoring the
overflow)

Example continued ...

ii) Using the incrementing machine, increase the above result 1000, in magnitude, by 1

1000

+1

1001 which is the same as obtained by ordinary subtraction.

Note

It may be noted that the above method of subtraction of binary numbers may be applied to subtraction of decimal numbers with the change that 9's complement of b will be added to a , instead in step (1). Following is the task in this regard

Equivalent machines

Two machines are said to be **equivalent** if they print the same output string when the same input string is run on them.

Remark: Two Moore machines may be equivalent. Similarly two Mealy machines may also be equivalent, but a Moore machine can't be equivalent to any Mealy machine. However, ignoring the extra character printed by the Moore machine, there exists a Mealy machine which is equivalent to the Moore machine.

Theorem

Statement:

For every Moore machine there is a Mealy machine that is equivalent to it (ignoring the extra character printed by the Moore machine).

Proof: Let M be a Moore machine, then shifting the output characters corresponding to each state to the labels of corresponding incoming transitions, machine thus obtained will be a Mealy machine equivalent to M .

Following is a note

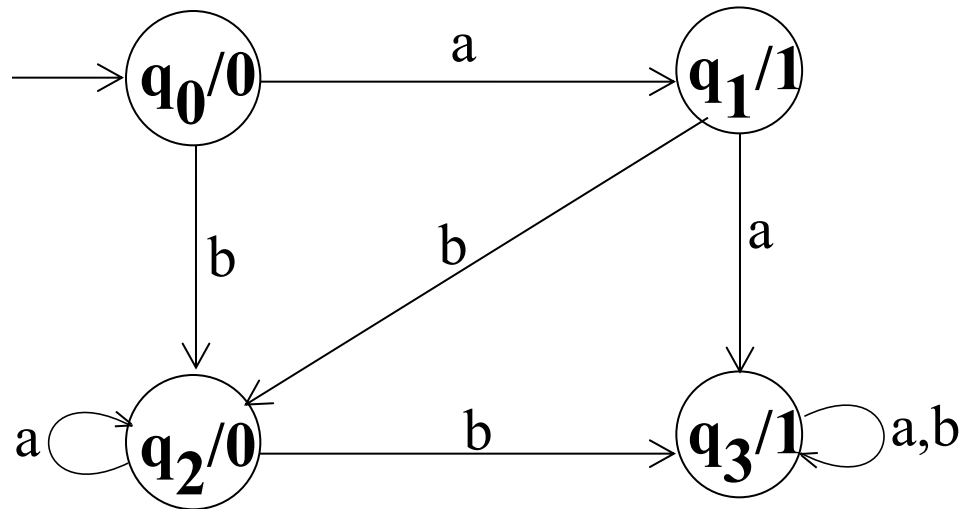
Note

It may be noted that while converting a Moore machine into an equivalent Mealy machine, the output character of a state will be ignored if there is no incoming transition at that state. A loop at a state is also supposed to be an incoming transition.

Following is the example of converting a Moore machine into an equivalent Mealy machine

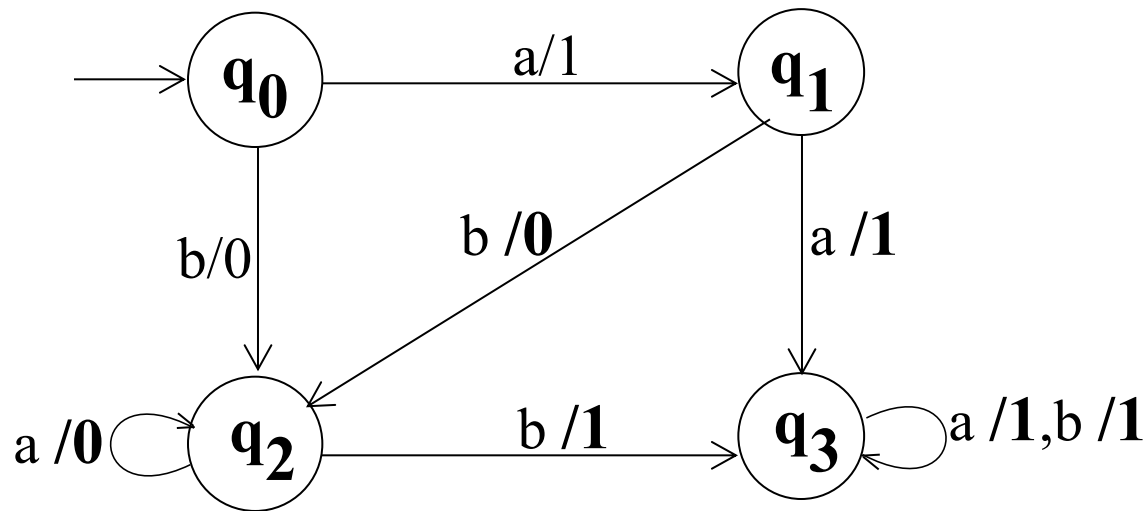
Example

Consider the following Moore machine



Using the method described earlier, the above machine may be equivalent to the following Mealy machine

Example continued ...



Running the string abbabbba on both the machines, the output string can be determined by the following table

Example continued ...

Input		a	b	b	a	b	b	b	a
States	q_0	q_1	q_2	q_3	q_3	q_3	q_3	q_3	q_3
Moore	0	1	0	1	1	1	1	1	1
Mealy		1	0	1	1	1	1	1	1

Theorem

Statement:

For every Mealy machine there is a Moore machine that is equivalent to it (ignoring the extra character printed the Moore machine).

Proof: Let M be a Mealy machine. At each state there are two possibilities for incoming transitions

1. The incoming transitions have the same output character.
2. The incoming transitions have different output characters.

Proof continued ...

If all the transitions have same output characters, then shift that character to the corresponding state.

If all the transitions have different output characters, then the state will be converted to as many states as the number of different output characters for these transitions, which shows that if this happens at state q_i then q_i will be converted to q_i^1 and q_i^2 *i.e.* if at q_i there are the transitions with two output characters then q_i^1 for one character and q_i^2 for other character.

Proof continued ...

Shift the output characters of the transitions to the corresponding new states q_i^1 and q_i^2 . Moreover, these new states q_i^1 and q_i^2 should behave like q_i as well. Continuing the process, the machine thus obtained, will be a Moore machine equivalent to Mealy machine M . Following is a note

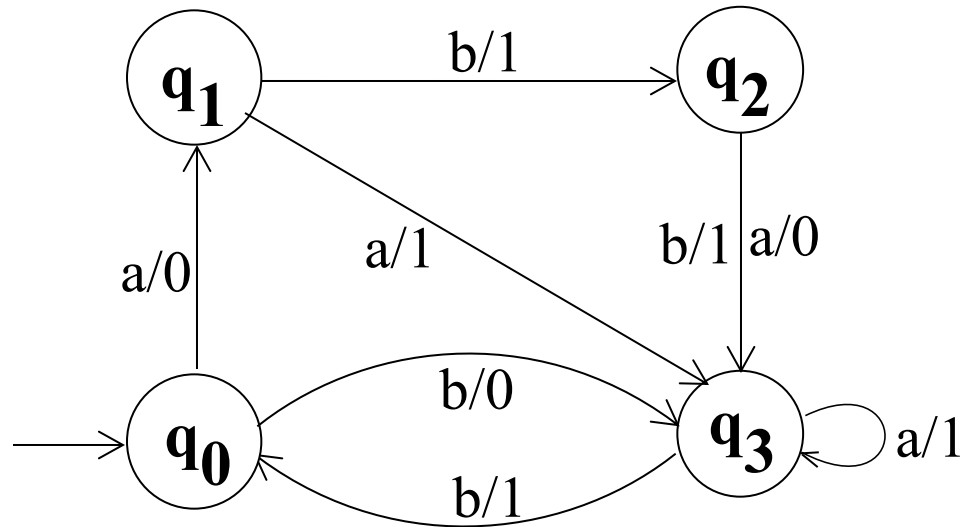
Note

It may be noted that if there is no incoming transition at certain state then any of the output characters may be associated with that state.

It may also be noted that if the initial state is converted into more than one new states then only one of these new states will be considered to be the initial state. Following is an example

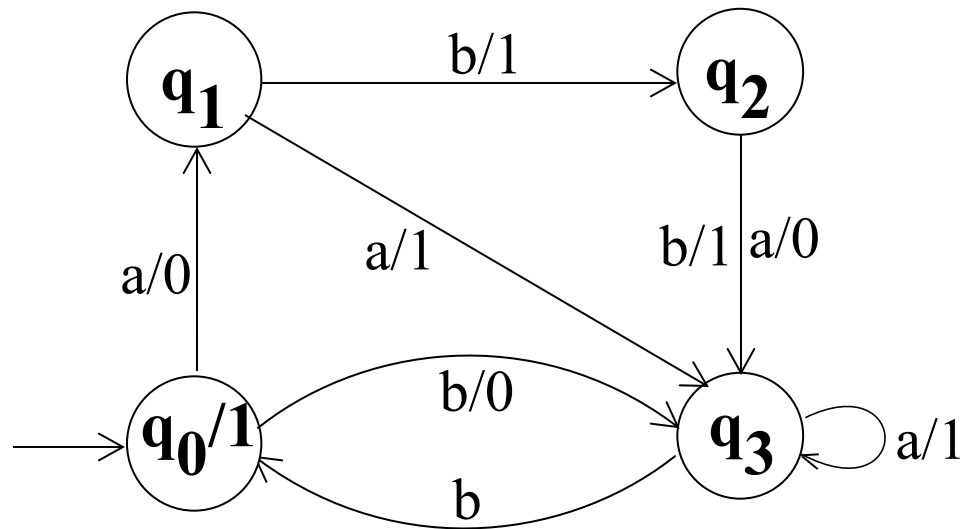
Example

Consider the following Mealy machine



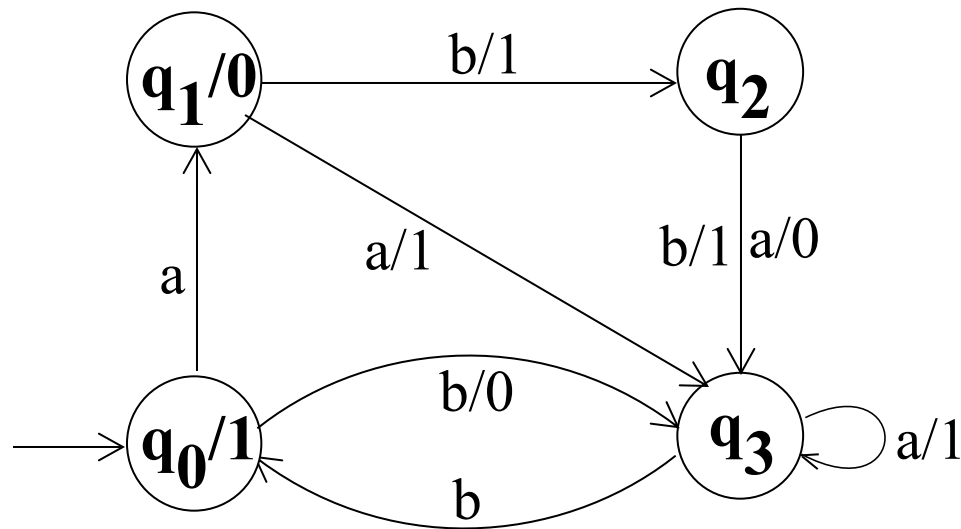
Example continued ...

Shifting the output character 1 of transition b to q_0



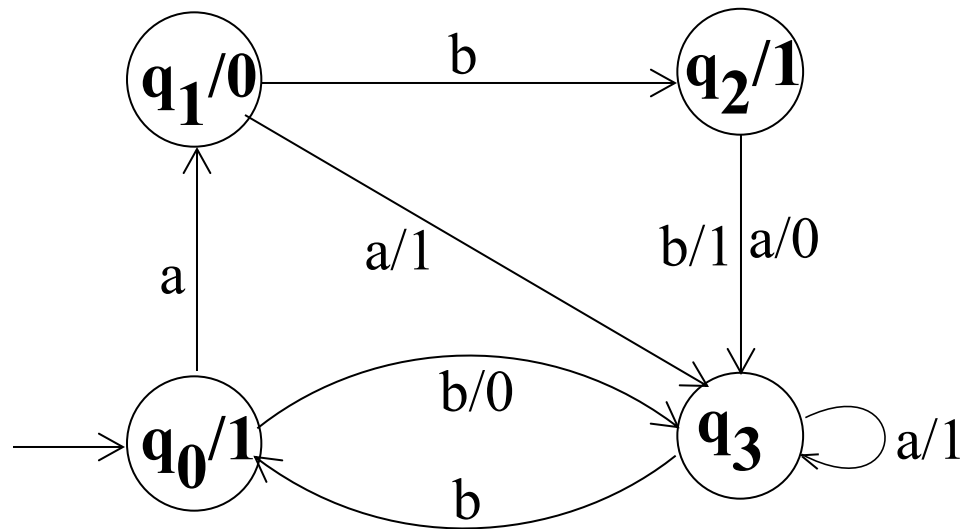
Example continued ...

Shifting the output character 0 of transition a to q_1



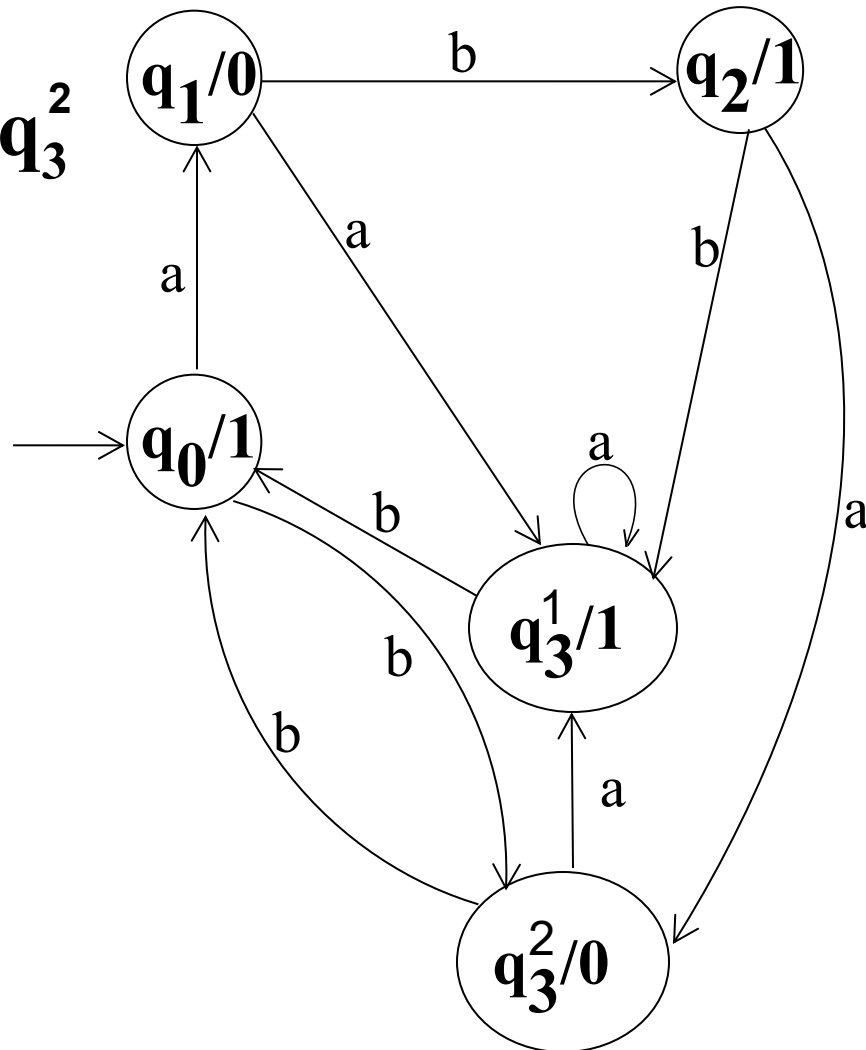
Example continued ...

Shifting the output character 1 of transition b to q_2



Example continued ...

Splitting q_3
into q_3^1 and q_3^2



Example continued ...

Running the string abbabbba on both the machines, the output strings can be determined by the following table

Input		a	b	b	a	b	b	b	a
States	q_0	q_1	q_2	q_3	q_3	q_0	q_3	q_0	q_1
Mealy		0	1	1	1	1	0	1	0
Moore	1	0	1	1	1	1	0	1	0

Summing Up

Applications of complementing and incrementing machines, Equivalent machines, Moore equivalent to Mealy, proof, example, Mealy equivalent to Moore, proof, example